Feature Analysis: Bornomala.com.bd vs Our Planned SaaS

> **Research Method:** Live browser exploration of `app.bornomala.com.bd/admin` (logged-in admin view for a real madrasa tenant — শ্যামিরখিল হামজা বিন আব্দুল মুত্তালিব দাখিল মাদ্রাসা).
>
> **Goal:** Map every feature Bornomala ships today, mark where we already cover it, and identify where we can leapfrog them by 10x.

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1. Bornomala Feature Inventory (What They Have Today)

1.1 Navigation Structure (main sidebar)

```
├── ড্যাশবোর্ড (Dashboard)
├── ভর্তি জিজ্ঞাসা (Admission Inquiry)
├── ছাত্র/ছাত্রী (Students)
│   ├── নতুন শিক্ষার্থী (New Student)
│   ├── সমস্ত শিক্ষার্থীবৃন্দ (All Students)
│   ├── Bulk Add Students
│   ├── Student Update All
│   ├── Student Shift (class/section transfer)
│   └── Class-wise views (Play → Class 10, Hifz)
├── উপস্থিতি (Attendance)
├── পরীক্ষা (Exam)
│   ├── পরীক্ষার রুটিন (Exam Routine)
```

```
│   ├── প্রবেশপত্র (Admit Card)
│   ├── আসন বিন্যাস (Seating Arrangement)
│   ├── মার্ক এন্ট্রি (Marks Entry)
│   └── নম্বর শীট ও ফলাফল কার্ড (Marksheet + Report Card)
├── সার্টিফিকেট (Certificate)
├── একাডেমিক ব্যবস্থাপনা (Academic Management)
│   ├── শ্রেণী (Class)
│   ├── শাখা (Section)
│   ├── সাবজেক্ট (Subject)
│   ├── শ্রেণী-সাবজেক্ট অ্যাসাইন
│   ├── শিক্ষক-সাবজেক্ট অ্যাসাইন
│   ├── পরীক্ষার সময়সীমা
│   ├── খরচের প্যাকেজ (Cost Package)
│   ├── চাঁদা সংগ্রহ (Donation Collection)
│   └── চাঁদা গ্রহণ (Donation Receive)
├── নোটিশ ব্যবস্থাপনা (Notice Management)
├── ফি সংগ্রহ ব্যবস্থাপনা (Fee Management)
│   ├── ফি সংগ্রহ (Fee Collection)
│   ├── আদায়কৃত ফি সমূহ (Collected Fees)
│   ├── ফি বরাদ্দ (Fee Allocation)
│   └── Cost Package
├── আয় সংগ্রহ (Income Collection)
```

```
├── ব্যয় (Expense)
├── বিনিয়োগ (Investment)
├── রিপোর্টস (Reports)
├── সেটিংস (Settings)
│   ├── কোম্পানি/ব্র্যান্ড (Company/Brand)
│   └── ডেটা টাইপ (Data Types)
├── শিক্ষক/কর্মী ব্যবস্থাপনা (Teacher/Staff Management)
├── স্টুডেন্ট রিলেশনশিপ ম্যানেজমেন্ট (SRM)
├── অভিভাবক (Parent/Guardian)
├── ব্যবহারকারী/ইউজারগণ (Users)
└── সেবা সমূহ (Services)
```

1.2 Fee Heads Supported

Tuition Fee, Tuition Fee Due, Admission Form, Admission Fee, Admission Fee Due, Session Fee, Session Fee Due, Exam Fee (1st/2nd/3rd/Final Term), Registration Fee, Package Fee, Transport Fee, Fine, ID Card Fee, Online Charge, Food Expense, Testimonial Fee, Sports Fee, Computer Lab Fee, Practical Fee, TC Fee, Syllabus Fee, Certificate Fee, Books Fee, Note Book Fee, Stationeries Fee, Development Fee, Dress Fee.

1.3 Expense Heads Supported

Admin Personal, Advance Payment, Asset, Bank Deposit, Bonus Pay, Voucher Bill, Electricity Bill, Entertainment, General Purchase, Honorarium Pay, Internet Bill, Investment Return, Labour Bill, Loan Paid, Office Expense, Office Rent, Overtime Bill, Product Purchase, Salary Pay, Software Bill, Stationery, Total Salary, Transport, Others.

1.4 Cash Receive Types

Admin Receive, Boss Personal, Bank Withdraw, Loan Receive, Loan Return, Opening Balance, Project Investment, Others.

1.5 Report Types (13)

Fee Collection, Fee Item, Tuition Fee Summary, Total Summary, Tuition Fee Due, Fee Report, Admission + Session Fee, Sales Category, Purchase, Expense, Cash Sheet, Cash Receive, Received & Payment Summary.

1.6 Per-Student Actions Observed

Ledger view, Print Admission Details, Admission Invoice, Edit, Student Shift (transfer to another class/section).

1.7 Other Observed Capabilities

- SMS module with balance tracking + SMS history + per-type counters (Attendance SMS, Notice SMS, Others)
- Invoice generation (`/add-invoice`)
- Support tickets system (`/support-tickets`)
- Multi-user role system (admin / user / teacher)
- Print workflows for admit card, admission form, marksheet

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2. Honest Assessment — What Works, What Doesn't

2.1 Bornomala's Strengths (things we MUST match)

1. **Deep fee taxonomy** — 27+ fee heads and 24+ expense heads cover real Bangladeshi institutional accounting. Principals trust them because every line item they ever need is already there.
2. **Per-student ledger** — one-click ledger view that shows every transaction against a student. This is loved by accountants.
3. **Admission Inquiry flow** — leads funnel before admission. Often overlooked.
4. **Class-wise student navigation in sidebar** — quick access pattern principals like.
5. **Donation Collection/Receive (চাঁদা)** — specific to madrasas and community-funded schools. Critical for Islamic institutions.
6. **Complete exam workflow** — routine → admit card → seating arrangement → marks entry → marksheet → result card is a solid end-to-end pipeline.
7. **Cash Flow and Investment tracking** — goes beyond fee collection into basic institutional accounting.
8. **Bulk student add / student shift** — saves hours during sessions.

2.2 Bornomala's Weaknesses (where we can leapfrog)

Technical / UX layer

- **Dated Bootstrap UI** — no glassmorphism, no micro-interactions, no skeleton loaders, no animations. Feels like 2015.
- **Legacy SutonnyMJ / Bijoy font** imported in CSS — breaks on copy/paste, breaks accessibility, breaks search. We're Unicode-first from day one.
- **No dark mode.**
- **No live/realtime updates** — everything is a full page reload.
- **No mobile app or PWA** — mobile browsing is clunky.
- **No offline mode** — fatal for Bangladeshi internet reality.
- **Dashboard is static cards** — no live charts, no trends, no AI insights. (Our demo already beats this.)
- **No keyboard shortcuts, no command palette.**
- **Page-level permissions only** — no granular per-class, per-subject, per-student scope control.

Functional gaps

- **No online payment integration** (bKash/Nagad/Rocket/SSLCommerz) — fees are cash/manual only. **Massive gap.**
- **No parent mobile-first portal** — SRM is admin-side.
- **No student portal** at all.
- **No push notifications / WhatsApp** — SMS-only, expensive.
- **No biometric or RFID attendance.**
- **No QR-based attendance or ID cards.**
- **No LMS features** — no homework submission, no online assignments, no grading.
- **No e-library / digital content delivery.**
- **No video conferencing integration.**
- **No library, hostel, transport modules** (or very basic).
- **No GPS transport tracking.**
- **No canteen management.**
- **No inventory (uniforms, stationery, books).**
- **No health/medical records.**
- **No counseling / disciplinary log.**
- **No scholarship management workflow.**
- **No PTA / parent community features.**
- **No alumni tracking.**

Madrasa-specific

- Generic fee/exam structure **does not track Hifz progress (para/sipara)**, **Kitab curriculum**, **Naazera/Tajweed stages**, or **Sabaq–Sabqi–Manzil** daily recording. These are core needs for madrasas.
- No Hijri calendar.
- No Arabic RTL layout option for madrasa reports.

Platform / scale

- **No API / webhooks** for integrations (payroll, govt reporting, SMS gateways).
- **No audit log visible.**
- **No 2FA.**
- **No multi-branch support** (chain madrasas with 2–5 campuses).
- **No data export backup** visible to admins.
- **Only Bengali UI** — no English, no Arabic.
- **No AI features** — no trend forecasting, no anomaly detection, no auto-generated report card comments, no dropout risk prediction.

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3. Feature Parity Matrix

#	Feature	Bornomala	Our Plan	Gap Status
1	Multi-tenant SaaS	✅	✅	Covered
2	Student admission (single)	✅	✅	Covered
3	Student admission (bulk Excel)	✅	✅	Covered
4	Admission inquiry/lead funnel	✅	❌	**Add to plan**
5	Student ledger (per-student)	✅	❌	**Add to plan**

#	Feature	Bornomala	Our Plan	Gap Status
6	Student transfer/shift	✅	❌	**Add to plan**
7	Attendance (daily)	✅	✅	Covered
8	Exam routine	✅	✅	Covered
9	Admit card generation	✅	✅	Covered
10	Seating arrangement	✅	❌	**Add to plan**
11	Marks entry	✅	✅	Covered
12	Marksheet / Report card PDF	✅	✅	Covered
13	Certificate generation	✅	❌	**Add to plan**
14	Class / Section / Subject CRUD	✅	✅	Covered
15	Class-Subject assignment	✅	✅	Covered
16	Teacher-Subject assignment	✅	✅	Covered
17	Fee head taxonomy (30+ types)	✅	⚠️ Generic	**Expand in plan**
18	Fee collection (cash/manual)	✅	✅	Covered
19	Fee allocation per class	✅	✅	Covered
20	Expense tracking (20+ heads)	✅	⚠️ Generic	**Expand in plan**

#	Feature	Bornomala	Our Plan	Gap Status
21	Cash flow report	✅	⚠️ Basic	**Expand in plan**
22	Investment tracking	✅	❌	**Add to plan**
23	Donation collection (চাঁদা)	✅	❌	**Add to plan (madrasa-critical)**
24	Invoice generation	✅	✅	Covered
25	Notice management	✅	✅	Covered
26	SMS module + history	✅	✅	Covered
27	Staff / employee management	✅	✅	Covered
28	Parent/guardian record	✅	✅	Covered
29	User/role management	✅	✅	Covered
30	Reports center (13+ types)	✅	⚠️ Basic	**Expand to 20+ report types**
31	Support tickets	✅	❌	**Add to plan**
32	Company/brand settings	✅	✅	Covered
33	**Online payment (bKash/Nagad/SSL)**	❌	✅	**We win 10x**
34	**Parent mobile portal**	❌	✅	**We win 10x**
35	**Student portal**	❌	✅	**We win 10x**

#	Feature	Bornomala	Our Plan	Gap Status
36	**PWA / offline mode**	❌	✅	**We win 10x**
37	**Live dashboard / realtime**	❌	✅	**We win 10x**
38	**Granular permissions**	⚠️ Coarse	✅	**We win 10x**
39	**Modern UX + dark mode**	❌	✅	**We win 10x**
40	**Unicode-first Bangla fonts**	❌ (SutonnyMJ)	✅	**We win**

Legend: ✅ Yes · ⚠️ Partial · ❌ No/Missing

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4. The 10x Differentiators (Our Killer Features)

These are features Bornomala does not offer that will immediately impress principals in a demo:

4.1 Tier 1 — Must-ship-day-one differentiators

1. **Online fee payment** — bKash, Nagad, Rocket, SSLCommerz. Parent taps, done. Bornomala has zero.
2. **Parent mobile PWA** — beautiful app-like experience; parents see attendance, results, dues, pay, message teacher.
3. **Student portal** — separate login, sees own data only, gamified attendance streaks, homework feed.
4. **Live realtime dashboard** — teachers take attendance and principal's ring-gauge updates instantly. (Already in our demo.)
5. **Granular scoped permissions** — class teacher ≠ subject teacher ≠ accountant. Principal grants per-class, per-subject, per-student access via checkboxes.
6. **Offline attendance + marks entry** — teacher marks attendance on shaky village wi-fi, syncs when online (IndexedDB + service worker).
7. **WhatsApp + SMS + Push** — not just SMS. Parents prefer WhatsApp.
8. **QR-based attendance & ID card** — student scans QR at gate, parent gets instant entry/exit SMS.

9. **Madrasa module** — Hifz para/sipara tracking, Kitab curriculum (Ibtedaiyyah → Fazilat → Kamil), Sabaq-Sabqi-Manzil daily log, Tajweed grade. None of this exists on Bornomala.
10. **Multi-branch / multi-campus** — one login manages 2–5 campuses with consolidated reporting. Chain madrasas (Jamia's with multiple branches) will buy instantly.

4.2 Tier 2 — Second-wave features (Phase 3–4)

11. **AI-generated report card comments** — teacher clicks "generate remark" and Claude writes a personalized paragraph per student.
12. **Dropout risk prediction** — ML model flags students with declining attendance + marks.
13. **GPS transport tracking** — parents see bus location live.
14. **Canteen management** — prepaid wallet, parent tops up, child eats, parent sees what was bought.
15. **Inventory** — uniforms, books, stationery with stock alerts.
16. **Health & medical record** — allergies, vaccinations, emergency contact.
17. **Counseling / disciplinary log** — private, principal-only visibility, audit-trailed.
18. **Scholarship workflow** — application → committee approval → fee auto-adjustment.
19. **Alumni tracking** — maintain lifetime connection for fundraising.
20. **PTA community** — parent groups, announcements, RSVP for events.
21. **e-Library** — digital textbooks, PDFs, past papers.
22. **Homework submission online** — student uploads, teacher grades, parent sees grade.
23. **Video classes** — Jitsi/Zoom/Google Meet integration, recording library.
24. **Command palette (⌘K)** — power users fly through the app.
25. **Keyboard shortcuts everywhere.**
26. **Customizable dashboard widgets** — principal rearranges their own view.
27. **API + webhooks** — integrate with govt reporting (EMIS, EIIN), payroll, accounting software.
28. **Audit log viewer** — "who changed this marks row?" answerable in 3 seconds.
29. **2FA + session management.**
30. **Automated backup + data export** — principal can download full data snapshot anytime (trust-builder).

4.3 Tier 3 — Platform plays

31. **Public school website per tenant** — `darululoom.bornomala.app` auto-generates a marketing site with admission form, news, gallery.
32. **Bangla/English/Arabic tri-lingual.**
33. **Hijri + Bengali + Gregorian calendars.**
34. **White-label / custom domain** (Enterprise).
35. **App Store marketplace** — third parties build plugins (biometric device drivers, extra report packs).

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5. Updates to PROJECT_PLAN.md

The following modules must be **added** to our existing plan:

5.1 New tables to add to database schema

```
-- Admission inquiry / lead funnel

admission_inquiries (id, school_id, student_name, guardian_phone,

  class_interested, source, status, followup_date, assigned_to, created_at)

-- Per-student financial ledger (derived view + snapshots)

student_ledger_entries (id, school_id, student_id, date, ref_type,

  ref_id, debit, credit, running_balance, note)

-- Donations (মাদ্রাসার চাঁদা)

donation_campaigns (id, school_id, title, target_amount, start_date, end_date)

donations (id, school_id, campaign_id, donor_name, donor_phone, amount,

  method, received_by, date, receipt_no)

-- Investments

investments (id, school_id, title, principal, return_expected,

  start_date, maturity_date, status, notes)

-- Exam seating

exam_rooms (id, school_id, name, capacity, rows, cols)

exam_seating (id, exam_subject_id, room_id, student_id, seat_row, seat_col)

-- Certificates (templates + issuance)

certificate_templates (id, school_id, type, html_template, variables JSONB)
```

certificates_issued (id, school_id, student_id, template_id, issued_on,

  serial_no, issued_by, pdf_url)

-- Support tickets

support_tickets (id, school_id, created_by, subject, body, priority,

  status, assigned_to, created_at)

support_messages (id, ticket_id, user_id, body, attachments JSONB, created_at)

-- Madrasa-specific

hifz_progress (id, school_id, student_id, para_no, sipara_no,

  status, tested_on, tested_by, mark, note)

kitab_curriculum (id, school_id, class_id, kitab_name, order)

daily_sabaq (id, school_id, student_id, date, sabaq, sabqi, manzil,

  quality, teacher_id)

-- Multi-branch

school_branches (id, school_id, name, address, is_primary, settings JSONB)

-- Transport, Library, Hostel (deferred but scaffold)

transport_routes / transport_vehicles / transport_students

library_books / library_issues

hostels / hostel_rooms / hostel_allocations

5.2 Fee-head seeding

Seed the 27 fee heads Bornomala uses by default so new tenants onboard instantly with the taxonomy they're used to.

5.3 Expense-head seeding

Seed the 24 expense heads as default categories.

5.4 Reports to build (expand from current plan)

Match Bornomala's 13 + add our own = **20 report types minimum**:

- Fee Collection, Fee Item, Tuition Summary, Tuition Due, Fee Report
- Admission/Session Fee Report, Donation Report
- Expense Report, Cash Sheet, Cash Receive Summary, Investment Return
- **+ New:** Attendance trend (30/90 day), Exam performance (class-wise, subject-wise), Teacher performance, Dropout risk, Dues aging, Daily Sabaq (madrasa), Hifz progress report, SMS usage, Audit log, Enrollment funnel (lead → admitted conversion)

5.5 New dashboard widgets to add

- Admission inquiry pipeline funnel (lead → visit → admitted)
- Donation campaign progress ring
- Hifz progress heatmap per student (for madrasas)
- Branch comparison bar (multi-campus)
- Overdue fees aging bucket (0-30 / 31-60 / 61-90 / 90+ days)

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6. Quick Competitive Positioning Pitch

When showing the demo to a principal:

> "Bornomala-এ যা আছে — সব আছে আমাদের। ২৭টা ফি হেড, ছাত্র লেজার, পরীক্ষার রুটিন, ফলাফল কার্ড, চাঁদা সংগ্রহ, ইনভেস্টমেন্ট — সব।
>
> তারপর আমরা যা দিচ্ছি যা ওদের নেই: — অভিভাবক bKash/Nagad-এ ফি দিবে, আপনাকে ক্যাশ গুনতে হবে না — অভিভাবকের মোবাইলে সুন্দর অ্যাপ, সন্তানের attendance live দেখবে — WhatsApp-এ notification, SMS খরচ ৭০% কমবে — মাদ্রাসার জন্য আলাদা মডিউল: হিফজের পারা, সবক-সবকী-মানজিল ডেইলি রেকর্ড — ইন্টারনেট না থাকলেও শিক্ষক attendance নিতে পারবে, পরে sync হবে — আপনার ২-৩টা শাখা থাকলে এক লগইনে সব ম্যানেজ করবেন — QR কোডে গেট এন্ট্রি, সাথে সাথে বাবা-মায়ের ফোনে SMS
>
> আর দাম? Bornomala-এর কাছাকাছি বা কম। কারণ আমরা cloud-first, automated, AI-driven।"

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7. Next Actions

1. Update `PROJECT_PLAN.md` to incorporate:
    - The 7 new tables above
    - Default fee-head and expense-head seeding
    - Admission Inquiry, Student Ledger, Student Shift as Phase 1 features (not Phase 2)
    - Donation module for madrasa tenants as Phase 1
    - Madrasa-specific Hifz tracking as a feature flag for madrasa-type tenants
2. Add the 10 Tier-1 differentiators to Phase 0–3 roadmap.
3. Keep Tier 2 features in backlog; prioritize by customer interviews.
4. Build a pricing comparison sheet (ours vs Bornomala) for the sales pitch.

Sources:

- Bornomala admin dashboard
- Students module
- Report center
- Public marketing site: bornomala.com.bd

End of comparison.